

Technical Report Overview:

Dataset and Problem Framing:

- This project uses the Telco Customer Churn dataset, which contains the data of 7,000 customers. The goal is to predict customer churn using features that include tenure, monthly charges, total charges, contract type, and service usage.

Data Processing and Assumptions:

- Some of the steps I used to make this data more usable included converting the TotalCharges column into a numeric format, filling in the missing values with the mean, and encoding the categorical variables using one-hot encoding.
- These steps allowed the data to be more interpretable and easier to work with while using the models. However, there are always a few limitations associated with every dataset, some of which include a broader customer population, which is not always the case. Important factors that would help determine churn, such as customer satisfaction and competitor pricing, are not included in this dataset.

Model Experiments:

- Linear Regression: The linear regression model was used to understand the basic relationships between the models and churn.
- Decision tree: Using the decision tree model helped to understand the more complex, non-linear relationships between features.
- Random forest: Predictive performance captures the relationship between the data by building multiple decision trees and combining their predictions.

```
Logistic Regression Accuracy: 0.7863733144073811

Confusion Matrix:
[[990  46]
 [255 118]]

Classification Report:
      precision    recall  f1-score   support

     0       0.80      0.96      0.87      1036
     1       0.72      0.32      0.44       373

   accuracy          0.79      1409
  macro avg       0.76      0.64      0.65      1409
 weighted avg       0.78      0.79      0.75      1409
```

```
Decision Tree Accuracy: 0.8076650106458482

Confusion Matrix:
[[966  70]
 [201 172]]

Classification Report:
      precision    recall  f1-score   support

     0       0.83      0.93      0.88      1036
     1       0.71      0.46      0.56       373

   accuracy          0.81      1409
  macro avg       0.77      0.70      0.72      1409
 weighted avg       0.80      0.81      0.79      1409
```

```
Random Forest Accuracy: 0.7352732434350603

Confusion Matrix:
[[1036   0]
 [ 373   0]]

Classification Report:
      precision    recall  f1-score   support

     0       0.74      1.00      0.85      1036
     1       0.00      0.00      0.00       373

   accuracy          0.74      1409
  macro avg       0.37      0.50      0.42      1409
 weighted avg       0.54      0.74      0.62      1409
```

Model Evaluations and Interpretations:

- Linear Regression: This model achieved an accuracy score of 78%, which makes it the worst-performing model in the analysis.

- The weak performance suggests a poor linear relationship among the factors that influence churn, such as contract type, tenure, and charges, with the outcome.
- This also indicates that predicting churn does need to involve complex comparisons with different features, making logistic regression not the best option for business use.
- **Decision tree:** The Decision Tree model achieved an accuracy of 80.8%, which makes it the best model in terms of accuracy.
- While this model helped add a non-linear split, the accuracy score suggests that this split was helpful to add predictive power for this dataset.
- This model did a great job at patching overall patterns in the dataset, meaning that customer churn in the dataset follows relatively clear patterns. This suggests that factors such as contract type, tenure, or service usage may strongly influence whether a customer decides to leave.
- **Limitations:** Decision trees can often overfit if the data is overloaded.
- **Random forest:** the random forest model achieved an accuracy score of approximately 73.5%, which is surprising considering that this is a lower score compared to the other two models.
- While this is a more complex model, it did not outperform the other models, meaning that the data I used did not contain any noisy patterns.
- Although this model achieved overall lower accuracy, it is designed to reduce overfitting by averaging multiple decision trees. This suggests that the model may be capturing more generalized patterns.
- **Limitations:** More complex models, such as the random forest model, do not always lead to better performance.

Limitations, Ethics, and Reflection:

- The goal of this project was to predict when a customer is likely to stop using a service (customer churn). While the models provided useful insights, they also have limitations.
- There is also potential for bias, where certain groups may be overrepresented, leading to skewed predictions. From an ethical perspective, using customer data requires responsibility; businesses must ensure privacy, maintain transparency, and avoid using predictions to target or disadvantage customers unfairly.
-

- Overall, the models can help identify customers at risk of leaving, supporting better retention strategies, but predictions must be interpreted carefully, considering their limitations, potential biases, and ethical implications.